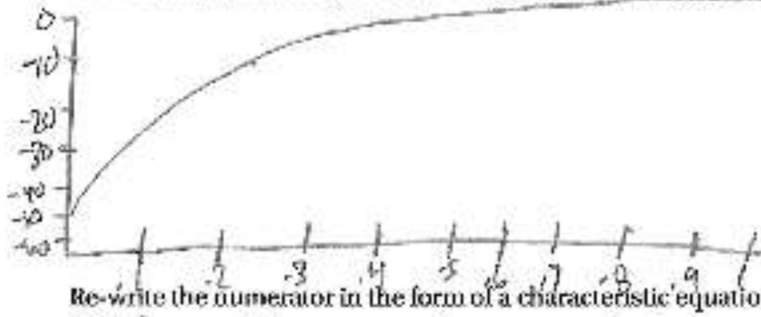
Draw the magnitude response for this system (use Matlab). What type of filter is this?



HPF

Re-write the numerator in the form of a characteristic equation. Solve for z.

$H(z) = .45 - 0.5z^{-1} + 0.05z^{-2}$

$(z^2)0 = .45 - 0.5z^{-1} + 0.05z^{-2}\,(z^2)$

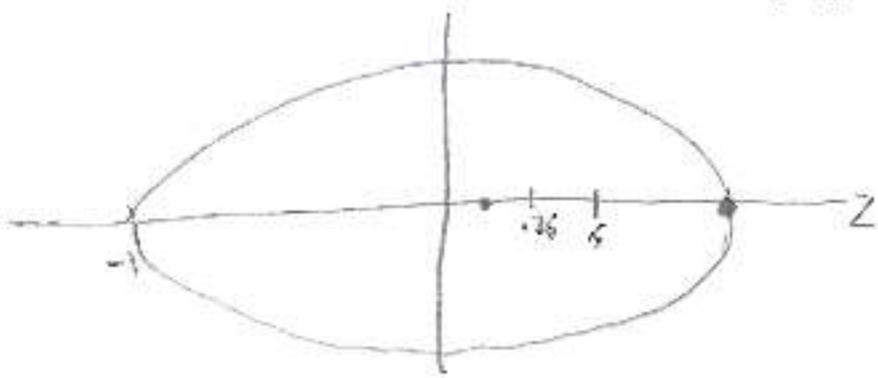
$0 = .45z^2 - 0.5z + .05$

Roots Function in Matlab ax^2+bx+c

$z = 1.00$ ← LPF

$z = .1111$

Plot the result for the numerator as 'zeros' within the unit circle on the complex plane.



4 Submission

To submit your homework, create a single zip file that contains the MATLAB function, along with the included sound file and GUI. Name the zip file: xxx_AET5420_HW3.zip, where xxx is your last name. Email the zip file to: eric.tarr@belmont.edu by 3:00 pm on April 1st.